



VIGNAN'S
Foundation for Science, Technology & Research
(Deemed to be UNIVERSITY)
-Estd. u/s 3 of UGC Act 1956

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UGC Entitled & AICTE Approved Online Degree Programs

**MCA with an elective in
Data Science**



Contact us

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MCA with elective in Data Science

Advanced Certification

- Full Stack Development
- Cloud Computing

About the elective

The data science sector in India is ranked among the fastest-growing industries, rising at the rate of 33%. Technology professionals that have experience in computer science, applications, and data science are in high demand. An individual who has good knowledge in using data to develop insights as well as integrating data science models into computer systems is priceless for companies.

MCA in Data Science is designed in a way to develop learners' computing skills and sharp analytical minds. The program gives a thorough understanding of tools needed to create end-to-end, data-powered software systems by laying strong foundations in software and computer applications.

Elective highlights

- Industry-focused curriculum that is designed by a panel of industry experts
- Learn how to compete and succeed on competitive data science platforms like Kaggle
- In-depth knowledge of all necessary technological tools such as SQL For ETL, NumPy, SciPy, ScikitLearn, Matplotlib, and Python
- Create a shareable e-portfolio of all your completed projects that demonstrate your skills
- Up-to-date career-specific content such as industry news, interview guides, resume guides, practice questions, and masterclasses

Learning Methodology

- 120 study hours in each course
- Interactive audio-video lectures
- Pre-recorded video lectures
- Discussion forum
- Self e-learning material
- Assignments, quizzes, MCQ, etc. for reinforcement
- Independent and group projects

Duration -
2 years (4 semesters)

Live Online Sessions -
Weekend

Advanced Certificate Program in Full Stack Development

Modern technology is so agile that developers can no longer grow if they are experts in just a single aspect of development. Developers today need to master the entire process of development - from design to real-time deployment. Most of the websites and apps that you use most frequently - Swiggy, Bookmyshow, Hotstar, Ola, etc. have all been built by full-stack developers in India and full-stack developers are among the most in-demand technology professionals in the industry today.

This trend has created an opportunity for developers to become Full Stack Developers. These developers work on all facets of development - frontend design, backend, database, debugging, testing and deployment. This advanced certification comprises of comprehensive learning on front end development with a course on Advanced Web Technologies, progress into building deeper understanding of the application logic with Application Development using Python course integrating with backend database by learning RDBMS and eventually learning CI/CD and DevOps will build skill sets in the area of deploying the application.

The Advanced Certificate Program in Full Stack Development will take you through an insightful journey in the following courses:

- Advanced Web Technologies
- Relational Database Management System
- Application Development Using Python
- CI / CD and DevOps

Advanced Certificate Program in Cloud Computing

Cloud computing has profoundly changed the way we live our lives and conduct business. Since COVID-19 pandemic, businesses have been quick when it comes to cloud

adoption, infrastructure, spending and development. According to leading research firm Cybersecurity Ventures, by the year 2025, over 50% of the total global data will be stored in the cloud. There will be over 100 zetta bytes of data stored in the cloud. (One zettabyte = one billion terabytes or one trillion gigabytes). Professionals with cloud computing expertise will be in high demand. Areas such as cloud security, database management and development, and cloud infrastructure management need professionals with a thorough understanding of cloud computing. Our advanced certificate program in cloud computing prepares learners to take on these opportunities with ease and advance their careers in the field of cloud computing and technology.

The Advanced Certificate Program in Cloud Computing will take you through an insightful journey in the following courses:

- Cloud Foundations
- Cloud Computing with AWS
- Cloud Managed Services
- Cloud Security and Migration

Eligibility Criteria:

All the graduates (Regular/Distance/Vocational) with a minimum of 50% of marks or 5.2 CGPA in any discipline or equivalent are eligible to apply. In the case of SC/ST graduates with a minimum of 45% of marks or 4.7 CGPA in any discipline or equivalent are eligible to apply.

Syllabus

1st
Semester

- Mathematical Foundation for Computer Application
- Operating System and Unix Shell Programming*
- Data Communication and Computer Networks
- Data Structures with Algorithms*
- Computer Organization and Architectures

2nd
Semester

- Relational Database Management System*
- Design and Analysis of Algorithms*
- Java Programming*
- Python for Data Science*
- Statistical Methods in Decision Making

3rd
Semester

- Application Development using Python*
- Advanced Web Technologies*
- Data Visualization*
- SQL for Data Science*
- Predictive Analytics using Machine Learning*
- Open Elective Course

4th
Semester

- Data Mining*
- Time series Analytics*
- Text Mining*
- Applied Analytics - Marketing, Web, Social Media
- Cross-Functional Elective Course
- Project**

* Courses which include Practicals (Lab Programs and Exercise)
** Project will be carried out between Sem 3 and Sem 4, but evaluation will reflect in Sem 4

Fee Structure:

For Indian National & SAARC Nations

Semester Fee Plan	₹ 27,500
Annual Fee Plan	₹ 55,000

Note

- 1. One Time University Registration Fee of ₹ 2500 is applicable during admission
- 2. Examination Fee of ₹ 4000 per year is applicable

For Foreign Students

Semester Fee Plan	\$550
Annual Fee Plan	\$1100

Note

- 1. One Time University Registration Fee of \$ 50 is applicable during admissionn
- 2. Examination Fee of \$ 60 per year is applicable

Process to Enroll:

- 1  Apply for your desired program
- 2  Upload documents
- 3  Pay tuition/program fee
- 4  Document verification

- 5  Provisional admission confirmation
- 6  Confirmation of admission
- 7  LMS activation



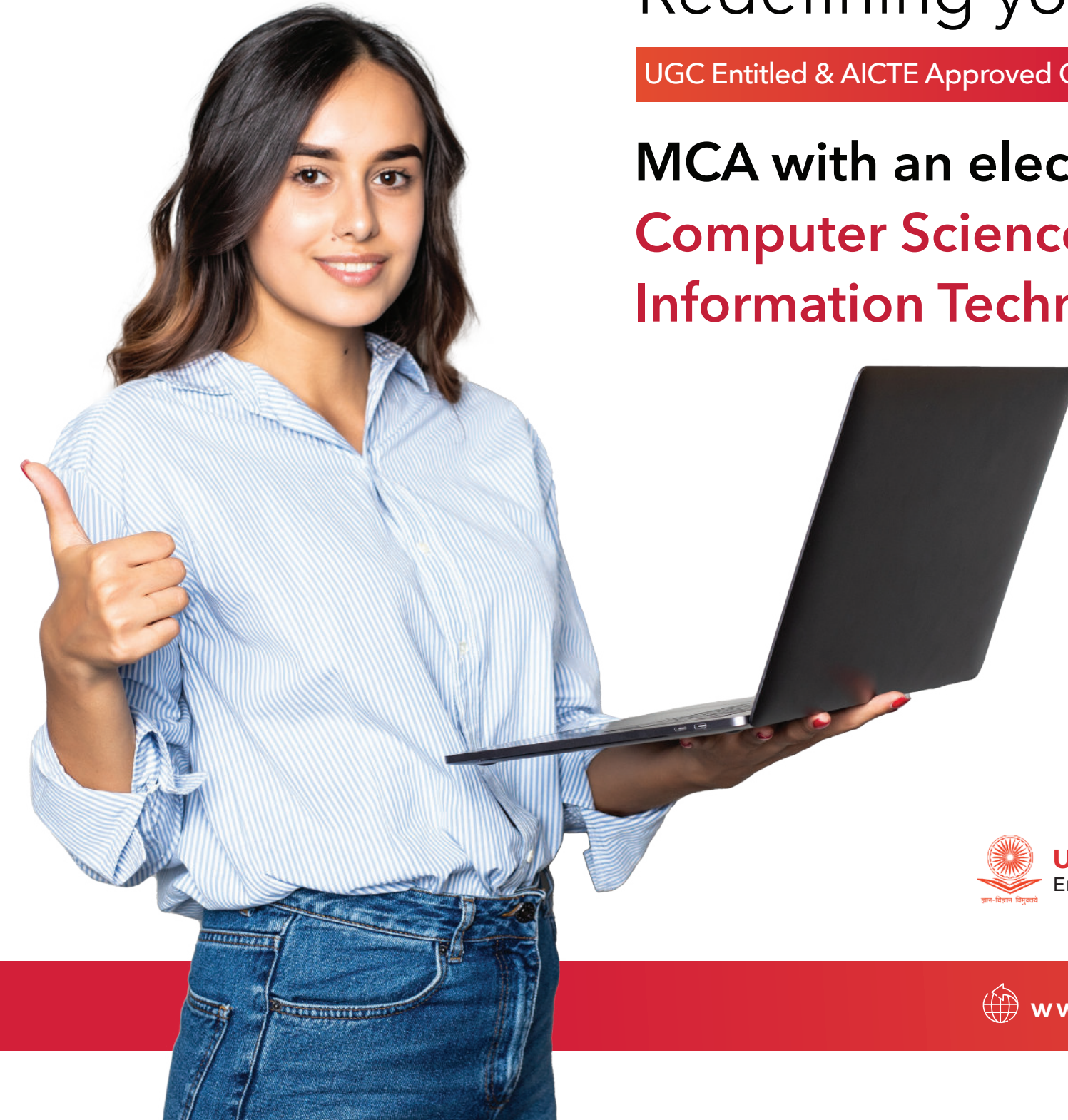
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MCA with elective in Computer Science and Information Technology

Advanced Certification

- Full Stack Development
- Artificial Intelligence and Machine Learning
- Cloud Computing

About the elective

An MCA Program in Computer Science and Information Technology is designed keeping in mind the growing need for competent professionals in the area of computer science and information technology. The program places a strong emphasis on the latest and the most popular programming languages and tools used for developing better software applications.

The program's curriculum aims to improve learners' technical knowledge and skills, giving them a head start on their careers for corporate roles in the software domain. It also covers every aspect of modern analytics, programming tools, techniques, and technologies used to address real-world challenges.

After the completion of an MCA in Computer Science and IT, a learner will be able to take up various job roles such as software engineer, data scientist, business analyst, web designer, and much more.

Elective highlights

- Builds a strong foundation in the critical areas of computer science and IT
- In-depth knowledge and practical exposure to programming, software engineering, web technologies, database management systems, etc
- Practical and lab assignments that go beyond the classroom
- Industry-relevant curriculum that offers technical excellence and enhances professional skills
- Become proficient in critical IT skills
- As a part of the program, design and develop projects that showcase your learnings

Learning Methodology

- 120 study hours in each course
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Advanced Certificate Program in Artificial Intelligence and Machine Learning

Artificial Intelligence (AI) and Machine Learning (ML) has taken the application of technology to levels never seen before. We, as consumers, prefer automated humanless personalized recommendations each time we start browsing the content on the internet. With about US\$ 50 billion spent on Artificial Intelligence, AI and its application is growing across businesses. The worldwide expenditure on Artificial Intelligence (AI) & Machine Learning (ML) is all set to grow exponentially in the next decade. This growth means an increased requirement of skilled employees across the world. Individuals with a strong foundation and hands-on experience with AI & ML applications will be in demand over the next few years. With the government's step towards digitization, and multiple organizations accelerating their digital transformation initiatives, the demand for AI talent pool is expected to skyrocket in India. This certification program prepares you for a rewarding career progression in the field of AI and ML.

The Advanced Certificate Program in AI & ML will take you through an insightful journey in the following courses:

- Probability and Statistics
- Artificial Intelligence
- Machine Learning
- Deep Learning
- CI / CD and DevOps

Advanced Certificate Program in Cloud Computing

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Operating System and Unix Shell Programming*
Data Communication and Computer Networks
Data Structures with Algorithms*
Computer Organization and Architectures

2nd
Semester

Relational Database Management System*
Design and Analysis of Algorithms*
Java Programming*
Network Security and Cryptography
Advanced Software Engineering

3rd
Semester

Application Development using Python*
Advanced Web Technologies*
Object Oriented Modeling and Design Patterns
Cloud Infrastructure and Services
Advanced Data Management Techniques*
Open Elective Course

4th
Semester

IT Project Management
Artificial Intelligence and Machine Learning*
Big Data Analytics
Internet of Things
Cross-Functional Elective Course
Project**

* Courses which include Practicals (Lab Programs and Exercise)
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